

Register Number

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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Computer Science and Engineering

Continuous Assessment Test – I

Regulations – R2021

Degree & Branch	BE - CSE				Semester	6
Subject Code & Name	UCS2604 - PRINCIPLES OF MACHINE LEARNING					
Academic Year	2023-2024 ODD /EVEN	Batch	2021-2025	Date	18.03.2024	FN / AN
Time: 08:15 - 09:45 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1	Explain the basic concepts of machine learning (K2)
CO2	Apply supervised algorithms for different classification problems (K3)
CO3	Explain the need for ensemble methods (K2)
CO4	Apply unsupervised and reinforcement learning techniques to various problems (K3)
CO5	Apply dimensionality reduction and optimization techniques (K3)

Part – A (4 × 2 = 8 Marks)

		KL	CO	PI
1.	Compare classification and regression problems with suitable examples.	K2	CO1	1.4.1
2.	Summarize Curse of Dimensionality with a suitable graph	K2	CO1	1.4.1
3.	Consider the company's performance over a five-year period. The closing price of stocks for these five years are given below. \$74.01 \$74.77 \$73.94 \$73.61 \$73.40 Apply the method of sum of squares and infer whether the customers could invest in the company's stocks or not.	K3	CO2	1.1.1 2.1.3
4.	Outline the need for kernel trick in Support Vector Machines.	K2	CO2	2.1.3

Part – B (3×6 = 18 Marks)

		KL	CO	PI
5.	Consider a binary classifier which classifies the corona tests of individuals as infected or non-infected. Given a dataset of 13 individuals which includes both infected and non-infected categories. The dataset has 8 infected and 5 non-infected individuals, where infected belong to class 1 and non-infected belong to class 0. The actual and predicted class labels are shown below. Actual Class = [1,1,1,0,0,0,1,0,0,0,0] Predicted Class = [0,0,0,1,1,1,1,0,0,0,0] Construct a confusion matrix and calculate precision, recall and F1 measure.	K3	CO1	1.1.1 2.1.3 13.3.1
6.	Interpret the need of bias, activation function and learning rate in training a neural network. Elaborate your answer on each parameter.	K2	CO2	1.4.1
7.	Outline the working of lazy learners. Illustrate the K-Nearest Neighbor Algorithm with necessary examples and diagrams.	K2	CO3	1.4.1 13.3.1

Part – C (2 × 12 = 24 Marks)
(Any subdivisions should be either 7+5 or 8+4)

		KL	CO	PI																		
8.	Consider a neural network with 2 inputs and 1 output. The output is '+1' (Plus 1) when there are even numbers of zeros or ones in the input. The output is '-1' (minus 1) when there are odd numbers of zeros or ones in the input. Model this network using a simple Perceptron Learning Algorithm and update weights when the output is misclassified. Note: Initialization values: w0 = w1 = w2 = 0.5, Learning Rate= 0.1, bias = -1, Wtheta = 1.0. Find the updated weight of each sample for one epoch. W_{new} = W_{old} - η(y_j-t_j). x_i Where, η is learning rate, y is predicted output, t is target and x is the vector of inputs.	K3	CO2	2.4.1 13.3.1																		
(Or)																						
9.	Use the given data to compute the correlation coefficient and build a regression model for finding the weight of a person with the input as height of a person. <table border="1"><tr><td>Height (X)</td><td>70</td><td>62</td><td>65</td><td>60</td><td>62</td><td>72</td><td>68</td><td>66</td></tr><tr><td>Weight (Y)</td><td>215</td><td>180</td><td>185</td><td>160</td><td>162</td><td>220</td><td>190</td><td>185</td></tr></table> If the height is 63 inches, can you predict the weight of this person using the model you built?	Height (X)	70	62	65	60	62	72	68	66	Weight (Y)	215	180	185	160	162	220	190	185	K3	CO2	2.4.1 13.3.1
Height (X)	70	62	65	60	62	72	68	66														
Weight (Y)	215	180	185	160	162	220	190	185														
10.	Apply Multi-Layer Perceptron algorithm for the given neural network and backpropagate the errors to reduce misclassification rate. Consider two inputs: 2 and 3. Target output :1, weights: w1 = 0.11, w2 = 0.21, w3 = 0.12, w4 = 0.08, w5 = 0.14 and w6 = 0.15 learning rate:1, Activation function: sigmoid. Calculate the output of the model in the forward pass. Compute the residue and back	K3	CO2	2.4.1 13.3.1																		

	propagate the error, update the weights. Perform one forward and one backward pass. (6 +6)			
(Or)				
11.	Consider a binary classification problem which separates data points into classes using a straight line. The distance between the straight line and the data points should be maximized. Identify the suitable classification algorithm for this specification and model using a constrained optimization problem by applying linear algebra. (5) Derive the function for the model which tolerates and does not tolerate errors. (4) Extend the model function to cater for non-linearly separable data. (3)	K3	CO2	2.4.1 13.3.1
